[20] Seat No: ___________ No. of printed pages :02

SARDAR PATEL UNIVERSITY

BCA (4th - Semester) Examination

Code No: US04CBCA52 : Data Structure - II

Date : 12-7-2024, Friday, Time :10:00 a.m. to 1:00 p.m.

Total Marks :[70]

Note : Right hand figure indicates the marks of each question.

Q.1 Multiple Choice Questions **[10]**

(1) __________Of tree is the maximum level of any node in the tree.

(a) Depth (b) Height (c) Width (d) None of these

(2) __________is a set of disjoint trees.

(a) Path. (b) Directed tree. (c) Forest. (d) Height.

(3) The number of sub tree of a node in a tree is called of a node.

(a) Degree. (b) Level. (c) Height. (d) Index.

(4) A linked list is_____________type of data structure.

(a) Linear (b) Non-Linear

(c) Both (A) and (B) (d) None of the Above

(5) _____________of the following is NOT the type of Linked list.

(a) One-way list (b) Doubly Linked list

(c) Three-way list (d) Circular linked list

(6) Each node in a linked list must contain at least _____________

(a) Three fields (b) Two fields

(c) Four fields (d) Five fields

(7) _________is the process to find the particular element from the array.

(a) Sorting (b) Inserting

(c) Searching (d) Updating

(8) _______________ design technique is used for Quick Sort.

(a) Divide and Conqueror (b) Dynamic Programming

(c) Greedy (d) Backtracking

(9) The number of records in a bucket is called the____________________.

(a) Bucket capacity (b) File Capacity

(c) Table capacity (d) None of these

(10) __________describes the storage of records on the tracks of a cylinder.

(a) Normal entry (b) Track index

(c) Prime entry (d) Master index

P.T.O

Q.2 Short Questions [Attempt any Ten] **[20]**

(1) List out applications of free
(2) Define: (A) Node, (B) Root
(3) Define Traversals. List the types of binary tree traversal.
(4) Lists insert operations that can be performed on singly linked list.
(5) Explain doubly circular linked list in brief.
(6) Explain singly linked list in brief.
(7) List out application of sorting.
(8) List out Categories of Sorting. Define any one of them
(9) Define: (A) Internal Sort (B) External Sort
(10) List popular hashing functions.
(11) Explain Space and Key Space.
(12) Explain in brief overflow area.

Q-3 Do as directed.

A Define Traversal. List all traversal and discuss any one traversal with algorithm **[05]**

B Draw the binary tree for following expressions: **[05]**

In order	n1	n2	n3	n4	n5	n6	n7	n8	n9
Postorder	n1	n3	n5	n4	n2	n8	n7	n9	n6

OR

A A/B+C)*(C+D*F)+(J/F)*(G/H). Draw the binary tree and find the preorder and post order of a binary tree **[05]**

B Write down an algorithm for preorder traversal of a binary tree **[05]**

Q-4 Do as directed.

A Define Linked List. List and explain types of linked lists. **[05]**

B Write an algorithm to insert node at front position in singly linked list. **[05]**

OR

A Write an algorithm to insert node at end position in singly linked list. **[05]**

B Write an algorithm to delete node at beginning in singly linked list. **[05]**

Q-5 Do as directed.

A Write down the algorithm of bubble sort. **[05]**

B Write down the algorithm of Binary Search. **[05]**

OR

A Write down the algorithm of Merge sort. **[05]**

B Write down the algorithm of Linear Search. **[05]**

Q-6 Do as directed.

A Write a detail note on: Collision resolution techniques **[10]**

OR

A Write a detail note on structure and processing of index sequential file supported by IBM **[10]**

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